

Raghav Agarwal

Software Engineer

Sydney, NSW | raghagarwal@gmail.com

[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

Software engineer at Commonwealth Bank with Python performance, applied-AI and full-stack delivery experience. Builds data-processing tools, interpretable AI workflows and web applications.

Technical skills

Python, JavaScript, TypeScript, SQL | Polars, pandas, NumPy

React, Redux, Node.js, Express, MongoDB | REST APIs, Git, unit testing

OpenAI, retrieval-augmented generation, agent workflows, Langfuse

Engineering experience

Commonwealth Bank | Associate Software Engineer

March 2025 – Present | Sydney

- Optimised legacy SAS/pandas data processing with Python and Polars, achieving an approximately 90× speed improvement; received Quarterly MVP recognition.
- Contributed to a Retail Credit Decisioning assistant in active use, explaining credit screens and referrals through tool-augmented AI workflows.
- Built bidirectional translation between Polars rules and plain-English pseudocode, supporting readable rule explanations and updates back into code.
- Developed retrieval-augmented workflows with OpenAI, human review and Langfuse tracing to support interpretable AI-assisted analysis.
- Developing Product Concierge, in testing with CBA staff, and a separate Decline Agent, in development, to explain loan-decline reasons and possible next steps.

Axiom Technologies | Software Engineer

June 2024 – December 2024 | Sydney

- Built a replacement timesheet application with React and Redux, connecting the interface to Node.js REST APIs and MongoDB to replace an existing Xero workflow.

ABN AMRO Clearing Bank | Junior Developer

December 2021 – April 2022 | Sydney

- Developed Python and SQL tools for MiFID II reporting and multi-source ETL, using pandas and NumPy with unit tests and logging in a Linux environment.

Selected projects

E-invoicing application — Implemented a MERN application for a university team capstone: PDF-to-UBL conversion, external validation, GridFS document storage and email sharing. Repository includes Jest and Cypress tests.

Reliable transport over UDP — Built a Python sender and receiver with sequence numbers, acknowledgements, retransmission and simulated packet loss.

Education and additional experience

UNSW | Bachelor of Computer Science | September 2021 – September 2024

Full-Stack Development 97/100; Python 90/100. Sudoku: Best Individual Project and Voters' Choice. Founder and President, Active Thinkers Society (2022–2024).

Earlier enterprise IT support: Capgemini, nbn, Johnson & Johnson and SC Johnson (2021–2025, overlapping roles).